Deep Low-Rank and Piecewise-Smooth Constraint Tensor Model for Hyperspectral Anomaly Detection

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***Abstract*— Low-rank tensor decomposition and autoencoder (AE) have recently attracted much attention in the hyperspectral anomaly detection field. Although the low-rank tensor decomposition model is self-explainable, a low-rank prior may only partially match real data. In contrast, AE can automatically learn the discriminative features between anomaly and background, whereas AE is not self-explainable. To overcome these problems, a deep low-rank and piecewise-smooth constraint tensor model is proposed for hyperspectral anomaly detection in this article, which makes full use of the low-rank and piecewise-smooth properties of the background tensor in model-driven tensor-based methods and combines the advantage of data-driven AE methods at the same time. Specifically, the background is reconstructed through the AE network, and the tensor nuclear norm (TNN) and total variation (TV) regularization are designed on the network output to make it self-explainable to exploit the low-rank and piecewise-smooth properties of the background in the 3-D. At the same time, an alternating direction method of multipliers (ADMMs)-based approach is given to solve the proposed model, making the low-rank and piecewise-smooth background tensor estimation process and the network training process into two sub-problems. Then, the background can be approximately reconstructed under the constraint of latent priors, while the anomaly is reconstructed with significant reconstruction errors. Finally, the reconstruction errors indicate the anomalous degree. Experimental results validated on several real hyperspectral datasets demonstrate the effectiveness of the proposed algorithm.**

***Index Terms*— Autoencoder (AE), hyperspectral anomaly detection, low-rank tensor, total variation (TV) regularization.**

NOMENCLATURE

Notations	*Descriptions*
x	Scalar.
$\mathbf{X}$	Matrix.
$\mathbf{X}(i, j)$	(i, j)th element.
$\odot$	Hadamard product.
$\|\mathbf{x}\|_2$	$\sqrt{\sum_i x_i^2}$.
$\|\cdot\|_*$	Matrix nuclear norm.
$\mathbf{x}$	Vector.
$\mathcal{X}$	Tensor.
$\mathcal{X}(i, j, :)$	$(i.j)$th tube.
$\|x\|$	Absolute value.
$\|\mathcal{X}\|_1$	$\sum_{ijk}\|\mathcal{X}_{ijk}\|$.
$\|\mathcal{X}\|_F$	$\sqrt{\sum_{ijk}\mathcal{X}_{ijk}^2}$.

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I. INTRODUCTION

HYPERSPECTRAL image (HSI) achieves excellent spectral resolution via utilizing hundreds of narrow-band that vary continuously from infrared to ultraviolet [1], [2], [3], and can accurately describe the reflection spectrum of objects, thereby significantly improving the ability of object detection and classification. The unique advantages of HSI lead to various applications, for example, target detection [4], [5] and classification [6], [7]. Specifically, hyperspectral target detection can be broadly divided into two categories: matching target [8] and anomaly detection [9], according to whether there is accurate prior information about the target. This article discusses hyperspectral anomaly detection, which can be used in various fields, such as precision agriculture, mineral recognition, and military surveillance [9], [10], [11].

The so-called hyperspectral anomaly detection refers to finding those pixels whose spatial and spectral characteristics significantly differ from the surrounding pixels without accurate prior information about the background or anomaly. Therefore, hyperspectral anomaly detection faces many challenges, such as insufficient prior information on the anomaly and background, complex and changeable background, and unbalanced distributions of the background and anomaly. In the past decade, methods based on various principles have been proposed for hyperspectral anomaly detection. Reed and Yu [12], proposed a classical statistics-based method called Reed-Xiaoli (RX) for hyperspectral anomaly detection based on the assumption that the background pixels obey multivariate Gaussian distribution while anomaly pixels do not. In RX, the whole pixels are adopted to calculate the mean and covariance of the background, and the Mahalanobis distance is adopted to detect the potential anomaly pixels. A more adaptive local RX is proposed in [13], which uses a double-sliding window to calculate the mean and covariance. Based on the detection principle of RX, some other improved variants have also been proposed, i.e., the kernel-RX (KRX) in [14], the cluster KRX in [15], the weighted-RX in [16] and [17], the RX-segmented in [18], the multiple window-based RX in [19], [20], and [21], and so on. Besides, some collaborative representation and sparse representation methods are also proposed for hyperspectral anomaly detection in [22], [23], [24], [25], [26], and [27], which assume that the background pixels can be

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represented by a dictionary, while anomaly pixels cannot. Further, by projecting the spectral domain features on the transform domain to increase the separability of background and target, some fractional domain-based detectors are proposed in [28], [29], [30], and [31] for hyperspectral anomaly detection.

Drawing on the idea of low-rank and sparse matrix decomposition (LRSMD) [32], [33], [34], if the hyperspectral data are regarded as a 2-D matrix, the original hyperspectral data matrix can be decomposed into a low-rank background matrix and a sparse anomaly matrix [35], [36], this kind of method is called LRSMD-based methods. Among them, the robust principal component analysis (RPCA) [37] is the most common one [35]. Yet, this RPCA does not consider the noise in the raw HSI data, which may affect the anomaly detection performance. Regarding the noise effect, a method called LSMAD was proposed in [36]. There is an additional noise matrix in the decomposition of HSI data compared with the RPCA, and the Mahalanobis distance is adopted to detect potential anomaly pixels. Further, a novel LMSD-MOG was proposed in [38], assuming the noise follows a mixed Gaussian distribution. Besides the LRSMD-based methods, some low-rank representation (LRR)-based methods are also introduced to deal with the hyperspectral anomaly detection task [39], [40]. Based on the concept of LRR, a method called AHMID was proposed in [41] to construct an accurate background dictionary for the background pixels composed of different land covers. However, the above methods may only partially use the latent properties of the background, such as the combination of the low-rank and piecewise-smooth properties being ignored, which may decrease the detection performance. A novel method, GTVLRR, was proposed in [42], where the total variation (TV) and low-rank constraints were designed on the background matrix to exploit the low-rank and the piecewise-smooth properties of the background. However, it is worth emphasizing that the LRSMD-based and the LRR-based need to unfold the three-order HSI data into a matrix, which ignores the multi-dimensional structure of the raw HSI data.

Actually, HSI data can be considered as a third-order tensor with one spectral dimension and two spatial dimensions. However, the matrix-based methods mentioned above do not consider this multidimensional structure, which may degrade detection performance. Some tensor-based methods have been proposed to deal with anomaly detection tasks, e.g., methods based on tensor completion [43] and methods based on tensor decomposition (tucker decomposition based in [44] and [45], CANDECOMP/PARAFAC based in [46], TPCA in [47] and [48]), and a tensor-regression-based method called PCA-TLRSR was proposed in [49]. However, these tensor-based methods only partially exploit the potential information of the background existing in the 3-D, i.e., the combination of low-rank and piecewise-smooth properties is ignored, which may degrade their performance. An improved tensor-based method called PTA was proposed to use the prior information in [50]. However, the PTA does not fully utilize the low-rank property of the background in spatial dimensions and the piecewise-smooth property of the background in spectral dimension. To fully exploit the low-rank and piecewise-smooth properties of the background, a tensor-representation-based method called LARTVAD was proposed in [51], where the low-rank property of the background in the 3-D and the piecewise-smooth property of the spatial dimensions are taken into account. Further, a method called TRDFTVAD was proposed in [52], where the tensor ring decomposition with factors TV regularization model is adopted to fully exploit the low-rank and piecewise-smooth properties of the background in 3-D.

With the development of deep learning technology, some data-driven background reconstruction models have been proposed for hyperspectral anomaly detection by adopting different networks to reconstruct the background. Those methods supposed that the reconstruction error of the background obtained by the network is small, and the reconstruction error of the anomaly is significant [53], [54], since the proportion of anomaly pixels in HSI is tiny, most are background pixels, so the model cannot learn abnormal features well. Thus, the reconstruction error is regarded as an anomaly. The most common of these are autoencoder (AE)-based detectors. A novel unsupervised feature representation method called spectrum-constrained adversarial AE (SC_AAE) by incorporating the spectral constraint strategy into an adversarial AE (AAE) is proposed in [55], which is based on the features of HSI to obtain better representations represented by hidden nodes degree of distinction. Another manifold-constrained AE network (MC-AEN)-based method is proposed in [56], in which an AE network with the learned embedding manifold constraints is adopted to preserve the intrinsic structure of hyperspectral data. However, the above two AE-based methods cannot realize end-to-end detection since they rely on the preprocessing [55] and detector construction procedures [56] as described in [57]. To realize end-to-end detection, an autonomous hyperspectral anomaly detection network based on fully convolutional AE with an adaptive-weighted strategy to reduce the weights of reconstruction for the anomaly was proposed detection in [57]. A guided AE network was proposed to reduce the representation ability of an AE for anomaly detection through a guided module in [53]. To facilitate the AE to generate background distributions more exhaustively and further reconstruct the background adequately, the detectors combine the AE network and the generative adversarial network were proposed in [58] and [59] to increase the separability of background and anomaly, where the discriminator is added to the output of the decoder in [58] or the outputs of the encoder and the decoder in [59]. By transforming the spectral dimension into a fractional Fourier domain, a frequency to spectrum mapping generative adversarial network is proposed in [60] to enhance the separability of the background and anomaly in the fractional Fourier domain. Besides, some algorithms based on blind-spot networks are also adopted for hyperspectral anomaly detection [54], [61], aiming to reconstruct the background more accurately and improve the anomaly detection performance. However, the features learned in the networks mentioned above are not interpretable, and some latent properties of the background may not be fully exploited. To combine the model-driven prior and the advantages of data-driven methods, the DeepLR was proposed in [62], which

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adopted the AE network with adapting weight to reconstruct the background. A low-rank constraint is designed on the output of the network to exploit the low-rank properties of the background in the spectral dimension. However, DeepLR only designed low-rank regularization on the spectral dimension by straightening the 3-D cube into a matrix, ignoring the low-rank property of the background on the spatial dimension, and damaging the multidimensional structure of hyperspectral data simultaneously. Besides, another important background property is ignored, namely the piecewise-smooth property in the spectral and spatial dimensions.

To make full use of the advantage of data-driven methods to reconstruct the background while preserving the inherent multi-dimensional structure of the hyperspectral and fully exploiting the potential properties of the background in 3-D, i.e., low-rank and piecewise-smooth, this article proposed a deep low-rank and piecewise-smooth constraint tensor model for hyperspectral anomaly detection. First, the hyperspectral cube is decomposed into background and anomaly tensors. The same AE network with adaptive weights in [62] was adopted to reconstruct the background tensor. Different from [62], the input and output of the AE network in this article both retain the multidimensional structure of the hyperspectral to make full use of the multidimensional structure characteristics of the background. Then, the tensor nuclear norm (TNN) and three-dimensional TV (3DTV) regularizations are designed on the output of the network to guide the AE network train. Finally, a scheme based on the alternating direction method of multipliers (ADMMs) is proposed to update the involved variables and the network parameters with all prior properties considered. The proposed method uses the powerful feature learning of the AE network to reconstruct the background and the potential low-rank and piecewise-smooth properties of the background in the 3-D to guide the AE network training to reconstruct the background better, thereby improving the performance of anomaly detection. The main contributions of this work can be summarized as follows.

1) We found that the low-rank and piecewise-smooth properties of the background exist both in the spatial and spectral dimensions. Thus, we propose to use these properties to guide the deep learning network training to reconstruct the background better, therefore improving the anomaly detection performance.
2) We combine the model-driven tensor model with the data-driven deep learning network, using the AE network to reconstruct the background and designing TNN and 3DTV constraints on the network output. On the one hand, it protects the inherent multi-dimensional structure of HIS data and fully uses the powerful feature learning capabilities of deep learning. On the other hand, it uses the low-rank and piecewise-smooth properties of the background in three directions to guide network training, making the reconstructed background more accurate.
3) We propose an iterative algorithm based on the ADMMs for the proposed model, which allows the network parameters and other parameters to be updated independently during each iterative solution process. Finally, we verified the effectiveness of the proposed algorithm on several real datasets.

The rest of this article is organized as follows. Some definitions and notations are presented in Section II. The proposed model is presented in Section III, and the solution process based on ADMM is also presented in this section. In Section IV, simulation results conducted on several real datasets are presented to demonstrate the effectiveness of the proposed algorithm. Finally, a summary is given in Section V.

II. Tensor Nation

Some symbols and the corresponding meanings are listed in Nomenclature, which will be used in the following. The following are some definitions of tensor.

Definition 1 (Mode-k Unfolding): The mode-k unfolding $\mathbf{R}_{(k)}$ of an Nth order tensor $\mathcal{R} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is an $I_k \times \prod_{n \neq k} I_n$ matrix, which satisfies that $\mathbf{R}_{(k)}(i_k, t)$ is mapped by $\mathcal{R}(i_1, i_2, \ldots, i_N)$, where $t = 1 + \sum_{(n=1, n\neq k)}^{N} (i_n - 1) * J_n$ with $J_n = \sum_{r=1, r\neq k}^{n-1} n_r$. The corresponding operator and inverse operator are denoted as $\mathbf{R}_{(k)} = \text{unfold}_k(\mathcal{R})$ and $\mathcal{R} = \text{fold}_k(\mathbf{R}_{(k)})$, respectively.

Definition 2 (t-Product [63]): The t-product of two tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{Y} \in \mathbb{R}^{n_2 \times n_4 \times n_3}$, denoted as $\mathcal{X} * \mathcal{Y}$, is a tensor $\mathcal{T} \in \mathbb{R}^{n1 \times n_4 \times n_3}$ with tubes

$$\mathcal{T}(m, n, :) = \sum_{t=1}^{n_2} \mathcal{X}(m, t, :) \star \mathcal{Y}(t, n, :)$$

where $\star$ represents the circular convolution [64] of two tubes.

Definition 3 (Tensor Transpose [63]): The transpose of a three-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ can be obtained by transposing each of the frontal slices and then reversing the order of transposed frontal slices 2 through n_3.

Definition 4 (Orthogonal Tensor [63]): The tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an orthogonal tensor only if $\mathcal{X} * \mathcal{X}^{\mathrm{T}} = \mathcal{X}^{\mathrm{T}} * \mathcal{X} = \mathcal{I}$, where $\mathcal{I}$ is the identity tensor [63].

Definition 5 (t-svd [63]): Suppose $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is a real-value tensor, then it can be decomposed as

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^{\mathrm{T}}$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensor. $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f-diagonal tensor [63] where each of its frontal slices is diagonal, and the entries in $\mathcal{S}$ are the tensor singular values of $\mathcal{X}$.

Definition 6 (Tensor Nuclear Norm): For a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and its FFT along the third dimension $\bar{\mathcal{X}} = \text{fft}(\mathcal{X}, [], 3)$, the TNN of $\mathcal{X}$ denoted as $\|\mathcal{X}\|_{\text{tnn}}$ is defined as the average of the nuclear norm of all the frontal slices in $\bar{\mathcal{X}}$, i.e.,

$$\|\mathcal{X}\|_{\text{tnn}} = \frac{1}{n_3} \sum_{i=1}^{n_3} \left\|\bar{\mathbf{X}}^{(i)}\right\|_*$$

where $\bar{\mathbf{X}}^{(i)}$ denotes the ith frontal slice of $\bar{\mathcal{X}}$.

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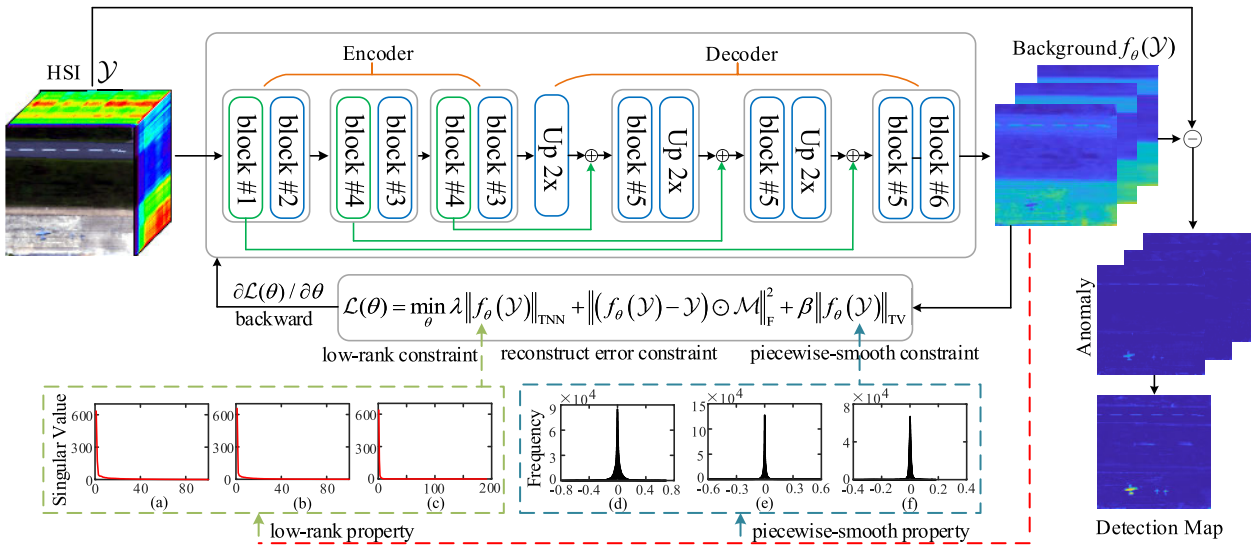


Fig. 1. Flowchart of the proposed model. (a)–(c) Denote the eigenvalues of the matrix $f_\theta(\mathcal{Y})_{(1)}$, $f_\theta(\mathcal{Y})_{(2)}$ and $f_\theta(\mathcal{Y})_{(3)}$, respectively. (d)–(f) Denote the static histograms of the pixel values of $\nabla_1 f_\theta(\mathcal{Y})$, $\nabla_2 f_\theta(\mathcal{Y})$ and $\nabla_3 f_\theta(\mathcal{Y})$, respectively. From (a)–(f), we know that the low-rank and piecewise-smooth properties exist in the spatial and spectral dimensions of the reconstructed background. Thus, we can design low-rank and piecewise-smooth regularizations on the output of the network to guide the network train.

III. Proposed Method

A. Problem Formulation and Motivation

From the perspective of hyperspectral data structure, it can be expressed as a third-order tensor composed of two spatial dimensions and one spectral dimension. Mathematically, it can be denoted as $\mathcal{Y} \in \mathbb{R}^{w\times h\times b}$, where w and h denote the width and height of the front two slices, and b is the number of spectral bands. In the existing tensor-based algorithms, it is generally assumed that the raw data tensor $\mathcal{Y}$ can be decomposed into a background tensor $\mathcal{B} \in \mathbb{R}^{w\times h\times b}$ and an anomaly tensor $\mathcal{A} \in \mathbb{R}^{w\times h\times b}$. Therefore, a general tensor decomposition-based model can be expressed as

$$\mathcal{Y} = \mathcal{B} + \mathcal{A}. \tag{1}$$

In the model-driven tensor models, some latent properties are exploited to improve the anomaly detection performance, i.e., the low-rank and piecewise-smooth properties of the background in the 3-D, the sparse of the anomaly. Finally, the tensor-based model can be generally constructed as

$$\begin{aligned} &\min_{\mathcal{B},\mathcal{A}} \Phi_{\text{lr}}(\mathcal{B}) + \lambda\Phi_{\text{ps}}(\mathcal{B}) + \beta\Phi_s(\mathcal{A}) \\ &\text{s.t. } \mathcal{Y} = \mathcal{B} + \mathcal{A} \end{aligned} \tag{2}$$

where λ and β are two non-negative parameters; $\Phi_{\text{lr}}(\cdot)$ and $\Phi_{\text{ps}}(\cdot)$ are the low-rank and piecewise-smooth constraint on the background tensor $\mathcal{B}$, respectively; and $\Phi_s(\cdot)$ is the sparse constraint on the anomaly tensor $\mathcal{A}$. However, due to the complexity of real data acquisition scenarios, the low-rank and piecewise-smooth priors may not exactly match the real data, leading to anomaly detection performance decreasing.

With the development of deep learning technology, some data-driven background reconstruction models are proposed to deal with hyperspectral anomaly detection tasks by exploiting the powerful feature learning ability of different networks. Researchers in the recent data-driven model generally believe the background can be reconstructed from the raw HSI data with a small reconstruction error. In contrast, the anomaly is reconstructed with a large reconstruction error. Finally, the reconstruction error is taken as the anomaly. Although this kind of data-driven model has powerful background reconstruction capabilities, almost all date-driven background reconstruction models are uninterpretable. Some important latent properties of the background used in the model-driven models, such as the low-rank and piecewise-smooth properties, are not interpretably utilized. In view of this, Wang et al. [62] combined the data-driven AE network with the model-based priors to propose a detector named DeepLR. This method makes full use of the low-rank property of the background in the spectral dimension by designing a low-rank regularization on the output of the AE network. The DeepLR model can be expressed explicitly as follows [62]:

$$\begin{aligned} &\min_\theta \frac{1}{2}\|(f_\theta(\mathbf{Z}) - \mathbf{Y})\mathbf{W}\|^2 + \lambda\|\mathbf{X}\|_* \\ &\text{s.t. } \mathbf{X} = f_\theta(\mathbf{Z}) \end{aligned} \tag{3}$$

where $\mathbf{Z} \in \mathbb{R}^{(w\times h)\times b}$ is the input uniform noise image of the network, $\mathbf{Y} \in \mathbb{R}^{(w\times h)\times b}$ is the input hyperspectral matrix data, $f_\theta(\mathbf{Z})$ is the output of the network with θ being the parameters of the network, $\mathbf{W}$ is the adaptive-weighted map. The AE network in [62] can reconstruct the background more precisely guided by model-driven low-rank prior, improving the anomaly detection performance to some extent.

However, the DeepLR model in [62] only exploits the low-rank property of the background in the spectral dimension, ignoring the low-rank property of the background in the spatial dimensions. Meanwhile, the piecewise-smooth property of the background existing in the spatial and spectral dimensions is ignored. Besides, the multidimensional structure of the hyperspectral data is destroyed since the 3-D cube is straightened into a matrix when adopting the matrix nuclear norm

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TABLE I
BLOCK STRUCTURES OF THE PROPOSED MODEL

block♯1	Conv2d(b-32,1x1,stride 1) BatchNorm2d(32) LeakyReLU	block♯4	Conv2d(32-32,1x1,stride 1) BatchNorm2d(32) LeakyReLU
block♯2	Conv2d(b-32,3x3,stride 2) BatchNorm2d(32) LeakyReLU Conv2d(32-32,3x3,stride 1) BatchNorm2d(32) LeakyReLU	block♯3	Conv2d(32-32,3x3,stride 2) BatchNorm2d(32) LeakyReLU Conv2d(32-32,3x3,stride 1) BatchNorm2d(32) LeakyReLU
block♯5	BatchNorm2d(64) Conv2d(64-32,3x3,stride 1) BatchNorm2d(32) LeakyReLU block♯4	block♯6	Conv2d(32-b,3x3,stride 1) Sigmoid

to exploit the low-rank property of the background in the spectral dimension. Those shortcomings of the DeepLR model may lead to anomaly detection performance degradation. In this article, inspired by the model-derived tensor model and the DeepLR model, a deep low-rank and piecewise-smooth constraint tensor model is proposed for hyperspectral anomaly detection (DLRPSAD), where the low-rank and piecewise-smooth properties of the background tensor in the 3-D are taken into account to guide the AE network train by designing the TNN and 3DTV regularization on the output of AE network. Thus, the proposed model can be mathematically expressed as

$$\min_{\theta} \|(f_\theta(\mathcal{Y}) - \mathcal{Y}) \odot \mathcal{M}\|_{\mathrm{F}}^2 + \lambda \|f_\theta(\mathcal{Y})\|_{\mathrm{tnn}} + \beta \|f_\theta(\mathcal{Y})\|_{\mathrm{TV}} \tag{4}$$

where $f_\theta(\mathcal{Y})$ and θ are the output of the network and the network parameters need to be trained, respectively, with the AE network structure being the same as that in [62] to make sure that the anomaly detection performance improvement of the proposed method is brought by the full use of the low-rank and piecewise-smooth properties of the background in the 3-D, and the detail block structures of the network is displayed in Table I; $\mathcal{M} \in \mathbb{R}^{w\times h\times b}$ is the adaptive-weighted tensor map; λ and β are two non-negative weights to balance the TNN and 3DTV regularization; $\|\cdot\|_{\mathrm{TV}}$ represents the 3DTV in this article. For a three-order tensor $\mathcal{Z} \in \mathbb{R}^{w\times h\times b}$, the 3DTV is expressed in detail as $\|\mathcal{Z}\|_{\mathrm{TV}} = \sum_{n=1}^{3} \|\nabla_n \mathcal{Z}\|_1$, where $\nabla_1 \mathcal{Z} \in \mathbb{R}^{(w-1)\times h\times b}$, $\nabla_2 \mathcal{Z} \in \mathbb{R}^{w\times(h-1)\times b}$ and $\nabla_3 \mathcal{Z} \in \mathbb{R}^{w\times h\times(b-1)}$ are the derivative tensors of $\mathcal{Z}$, which are calculated via

$$\begin{cases} \nabla_1 \mathcal{Z}(i,j,k) = \mathcal{Z}(i+1,j,k) - \mathcal{Z}(i,j,k) \\ \nabla_2 \mathcal{Z}(i,j,k) = \mathcal{Z}(i,j+1,k) - \mathcal{Z}(i,j,k) \\ \nabla_3 \mathcal{Z}(i,j,k) = \mathcal{Z}(i,j,k+1) - \mathcal{Z}(i,j,k). \end{cases} \tag{5}$$

The flowchart of the proposed DLRPSAD model is given in Fig. 1. From Fig. 1(a)–(c), we can know that the low-rank property does exist in the 3-D of the reconstructed background obtained by the AE network. Similarly, the piecewise-smooth property also exists in the 3-D of the reconstructed background obtained by the AE network. Thus, it is necessary to design low-rank and piecewise-smooth constraints on the output of the AE network. In view of this, the TNN and 3DTV regularization are designed on the output of the AE network in the proposed DLRPSAD model to fully exploit the low-rank and piecewise-smooth properties of the background in the spatial and spectral dimensions, i.e., the second and the third terms in (4), respectively. Overall, the proposed model combines the advantage of the powerful feature learning ability of the data-driven AE-based model and the efficient use of latent prior in the self-explainable model-driven tensor-based model. On the one hand, the proposed model fully protects the inherent multi-dimensional structure of hyperspectral data. On the other hand, the proposed method makes full use of the advantages of the data-driven and model-driven methods using the potential background prior information to guide the training of the AE network, which can better reconstruct the background tensor, thereby improving the performance of anomaly detection.

B. Optimization via ADMM

The ADMMs is adopted to solve the parameters in (4). Several auxiliary variables are introduced to simplify the solution process. Thus, (4) can be reconstructed as

$$\begin{aligned} &\arg\min_{\theta,\mathcal{X},\mathcal{W}_n} \|(f_\theta(\mathcal{Y}) - \mathcal{Y}) \odot \mathcal{M}\|_{\mathrm{F}}^2 + \lambda \|\mathcal{X}\|_{\mathrm{tnn}} + \beta \sum_{n=1}^{3} \|\mathcal{W}_n\|_1 \\ &\text{s.t.} \begin{cases} f_\theta(\mathcal{Y}) = \mathcal{X} \\ \nabla_n f_\theta(\mathcal{Y}) = \mathcal{W}_n, \quad n = 1,2,3 \end{cases} \end{aligned} \tag{6}$$

where $\mathcal{W}_1 \in \mathbb{R}^{(w-1)\times h\times b}$, $\mathcal{W}_2 \in \mathbb{R}^{w\times(h-1)\times b}$, $\mathcal{W}_3 \in \mathbb{R}^{w\times h\times(b-1)}$ and $\mathcal{X} \in \mathbb{R}^{w\times h\times b}$ are four auxiliary variables. Thus, the augmented Lagrangian function of (6) can be constructed as

$$\begin{aligned} &L(\theta, \mathcal{X}, \mathcal{W}_n, \mathcal{Q}, \mathcal{P}_n) \\ &= \|(f_\theta(\mathcal{Y}) - \mathcal{Y}) \odot \mathcal{M}\|_{\mathrm{F}}^2 + \lambda \|\mathcal{X}\|_{\mathrm{tnn}} \\ &\quad + \beta \sum_{n=1}^{3} \|\mathcal{W}_n\|_1 + \mu \|f_\theta(\mathcal{Y}) - \mathcal{X}\|_{\mathrm{F}}^2 + \langle \mathcal{Q}, f_\theta(\mathcal{Y}) - \mathcal{X}\rangle \\ &\quad + \sum_{n=1}^{3} \mu \|\nabla_n f_\theta(\mathcal{Y}) - \mathcal{W}_n\|_{\mathrm{F}}^2 + \langle \mathcal{P}_n, \nabla_n f_\theta(\mathcal{Y}) - \mathcal{W}_n\rangle \end{aligned} \tag{7}$$

where $\mathcal{Q} \in \mathbb{R}^{w\times h\times b}$, $\mathcal{P}_1 \in \mathbb{R}^{(w-1)\times h\times b}$, $\mathcal{P}_2 \in \mathbb{R}^{w\times(h-1)\times b}$ and $\mathcal{P}_3 \in \mathbb{R}^{w\times h\times(b-1)}$ are Lagrange multipliers and μ is a non-negative penalty parameter. Still now, the variables θ, $\mathcal{X}$, $\mathcal{Q}$, $\mathcal{W}_n$, and $\mathcal{P}_n$, $(n = 1, 2, 3)$ are each independent. In this way, the variables involved in (7) can be decomposed into several independent sub-problems to be solved. In particular, one of the variables can be updated by fixing the other variables as follows.

1) *Update $\mathcal{X}$:* The augmented Lagrangian function with respect to $\mathcal{X}$ can be expressed as

$$\begin{aligned} \mathcal{L}(\mathcal{X}) &= \arg\min_{\mathcal{X}} \lambda \|\mathcal{X}\|_{\mathrm{tnn}} + \mu^k \|f_{\theta^k}(\mathcal{Y}) - \mathcal{X}\|_{\mathrm{F}}^2 \\ &\quad + \left\langle \mathcal{Q}^k, f_{\theta^k}(\mathcal{Y}) - \mathcal{X}\right\rangle \\ &= \arg\min_{\mathcal{X}} \lambda \|\mathcal{X}\|_{\mathrm{tnn}} + \mu^k \left\| f_{\theta^k}(\mathcal{Y}) - \mathcal{X} + \frac{\mathcal{Q}^k}{\mu^k} \right\|_{\mathrm{F}}^2 \end{aligned} \tag{8}$$

whose solution can be obtained according to [65]. Define an intermediate variable as follows:

$$\tilde{\mathcal{X}}^k = f_{\theta^k}(\mathcal{Y}) + \frac{1}{\mu^k} \mathcal{Q}^k \tag{9}$$

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and computing $\tilde{\mathcal{X}}^k = \mathcal{U}^k * \mathcal{S}^k * \mathcal{V}^k$. Then, we have

$$\mathcal{X}^{k+1} = \mathcal{U}^k * \mathcal{S}^k_{\frac{\lambda}{\mu^k}} * \mathcal{V}^k \tag{10}$$

where

$$\begin{cases} \mathcal{S}^k_{\frac{\lambda}{\mu^k}} = \text{iff}\left(\bar{\mathcal{S}}^k_{\frac{\lambda}{\mu^k}}, [], 3\right) \\ \bar{\mathcal{S}}^k_{\frac{\lambda}{\mu^k}}(:,:,i_b) = \max\left(\bar{\mathcal{S}}^k(:,:,i_b) - \dfrac{\lambda}{\mu^k}, 0\right), \\ \qquad\qquad\qquad\qquad i_b = 1, \ldots, b \\ \bar{\mathcal{S}}^k = \text{fft}\left(\mathcal{S}^k, [], 3\right). \end{cases} \tag{11}$$

2) *Update $\mathcal{W}_n, (n = 1, 2, 3)$:* The augmented Lagrangian function with respect to $\mathcal{W}_n$ can be expressed as

$$\begin{aligned} \mathcal{L}(\mathcal{W}_n) &= \arg\min_{\mathcal{W}_n} \beta\|\mathcal{W}_n\|_1 + \mu^k\|\nabla_n f_{\theta^k}(\mathcal{Y}) - \mathcal{W}_n\|_{\text{F}}^2 \\ &\quad + \left\langle \mathcal{P}_n^k, \nabla_n f_{\theta^k}(\mathcal{Y}) - \mathcal{W}_n \right\rangle \\ &= \arg\min_{\mathcal{W}_n} \mu^k \left\| \nabla_n f_{\theta^k}(\mathcal{Y}) - \mathcal{W}_n + \frac{\mathcal{P}_n^k}{\mu^k} \right\|_{\text{F}}^2 \\ &\quad + \beta\|\mathcal{W}_n\|_1. \end{aligned} \tag{12}$$

Define

$$\tilde{\mathcal{W}}_n^k = \nabla_n f_{\theta^k}(\mathcal{Y}) + \frac{1}{\mu^k}\mathcal{P}_n^k. \tag{13}$$

Then, $\mathcal{W}_n$ can be solved via

$$\mathcal{W}_n^{k+1}(i_w^n, i_h^n, i_b^n) = \text{soft}_1\left(\tilde{\mathcal{W}}_n^k(i_w^n, i_h^n, i_b^n), \frac{\beta}{\mu^k}\right) \tag{14}$$

where $(i_w^1 = 1, \ldots, w-1)$, $(i_w^2/i_w^3 = 1, \ldots, w)$, $(i_h^2 = 1, \ldots, h-1)$, $(i_h^1/i_h^3 = 1, \ldots, h)$, $(i_b^3 = 1, \ldots, b-1)$, $(i_b^1/i_b^2 = 1, \ldots, b)$ and

$$\text{soft}_1(y, m) = \text{sign}(y)\max(|y| - m, 0) \tag{15}$$

where $\text{sign}(\cdot)$ is the sign function.

3) *Update θ:* The augmented Lagrangian function with respect to the network parameters $\boldsymbol{\theta}$ need to the updated can be expressed as

$$\begin{aligned} \mathcal{L}(\theta) &= \arg\min_{\theta} \|(f_\theta(\mathcal{Y}) - \mathcal{Y}) \odot \mathcal{M}\|_{\text{F}}^2 \\ &\quad + \mu^k \left\| f_\theta(\mathcal{Y}) - \mathcal{X}^{k+1} \right\|_{\text{F}}^2 + \left\langle \mathcal{Q}^k, f_\theta(\mathcal{Y}) - \mathcal{X}^{k+1} \right\rangle \\ &\quad + \sum_{n=1}^{3} \mu^k \left\| \nabla_n f_\theta(\mathcal{Y}) - \mathcal{W}_n^{k+1} \right\|_{\text{F}}^2 \\ &\quad + \left\langle \mathcal{P}_n^k, \nabla_n f_\theta(\mathcal{Y}) - \mathcal{W}_n^{k+1} \right\rangle \\ &= \arg\min_{\theta} \|(f_\theta(\mathcal{Y}) - \mathcal{Y}) \odot \mathcal{M}\|_{\text{F}}^2 \\ &\quad + \mu^k \left\| f_\theta(\mathcal{Y}) - \mathcal{X}^{k+1} + \frac{\mathcal{Q}^k}{\mu^k} \right\|_{\text{F}}^2 \\ &\quad + \sum_{n=1}^{3} \mu^k \left\| \nabla_n f_\theta(\mathcal{Y}) - \mathcal{W}_n^{k+1} + \frac{\mathcal{P}_n^k}{\mu^k} \right\|_{\text{F}}^2. \end{aligned} \tag{16}$$

The loss functions corresponding to the network parameters θ are all F-norm-based, so the gradients can be easily calculated in current deep-learning frameworks. We use adaptive moment estimation (Adam) to update θ, i.e., update $f_\theta(\mathcal{Y})$. Specifically, we employ one gradient descent step using the Adam in each update.

Algorithm 1 Proposed DLRPSAD Algorithm

Input: Tensor data $\mathcal{Y}$, parameters λ, β.
Initialize: Set the variables $\mathcal{X}$, $\mathcal{W}_n$, $\mathcal{P}_n$ as zero tensors, $\rho = 1.02$, $\mu_0 = 0.5$, $\mu_{\max} = 10^9$, $th = 10^{-4}$.
While ($\varepsilon > th$ and $k < k_{\max}$) **do**
1) Update $\mathcal{X}$ via eq(10).
2) Update $\mathcal{W}_n$ via eq(14),
3) Update θ via eq(16).
6) Update $\mathcal{P}_n$, $\mathcal{Q}_n$ $(n = 1, 2, 3)$, and μ via (17).
7) Judge the stop condition according to (20).
8) $k = k + 1$.
End while
Output Obtain $\bar{\mathbf{D}}$ according to (22) and (23).

4) *Update Multipliers:* The multipliers $(\mathcal{Q}, \mathcal{P}_n, n = 1, 2, 3)$ and the penalty parameter μ can be updated as follows:

$$\begin{cases} \mathcal{Q}^{k+1} = \mathcal{Q}^k + \mu^k\left(f_{\theta^{k+1}}(\mathcal{Y}) - \mathcal{X}^{k+1}\right) \\ \mathcal{P}_n^{k+1} = \mathcal{P}_n^k + \mu^k\left(\nabla_n f_{\theta^{k+1}}(\mathcal{Y}) - \mathcal{W}_n^{k+1}\right) \\ \mu^{k+1} = \min\left(\rho\mu^k, \mu_{\max}\right) \end{cases} \tag{17}$$

where $\rho > 1$ is a constant to reduce the impact of the initialization of μ, and $\mu_{\max}$ is the upper bound of μ.

5) *Update $\mathcal{M}$:* Similar to [62], each front slice of the adaptive-weighted map $\mathcal{M}$, denoted as $\mathcal{M}(:,:,i_b)$, $i_b = 1, 2, \ldots, b$, is calculated via

$$\mathcal{M}(i_w, i_h, i_b) = \max(\mathbf{E}^k) - e^k_{i_w, i_h} \tag{18}$$

where $\mathbf{E}^k \in \mathbb{R}^{w\times h}$ is a matrix with the (i_w, i_h)th, $(i_w = 1, \ldots, w, i_h = 1, \ldots, h)$ element being $e^k_{i_w,i_h}$, and $e^k_{i_w,i_h}$ is calculated via

$$e^k_{i_w,i_h} = \|f_{\theta^k}(\mathcal{Y})(i_w, i_h, :) - \mathcal{Y}(i_w, i_h, :)\|_{\text{F}}^2. \tag{19}$$

In this article, the adaptive-weighted map $\mathcal{M}$ is update every 50 iterations according to (18), which is similar to the updating strategy in [62].

6) Similarly, the stop converges adopted in this article are the same as in [62]. Specifically, the training of the proposed model stops when the average loss ε is below the set threshold th within the last 50 iterations, which is judged via

$$\varepsilon = \sum_{i=k}^{k+50}\left(\mathcal{L}(\theta^{k+1}) - \mathcal{L}(\theta^k)\right) < \text{th} \tag{20}$$

where k denotes the number of iterations. The stop threshold th is set as 0.0001 in this article.

When the network training reaches the convergence condition, the last output of the network, denoted as $f_{\theta^l}(\mathcal{Y})$, is regarded as the reconstructed background tensor. Then, the anomaly tensor can be calculated via

$$\mathcal{A} = \mathcal{Y} - f_{\theta^l}(\mathcal{Y}). \tag{21}$$

Up to now, the anomaly detection map, denoted as $\mathbf{D} \in \mathbb{R}^{w\times h}$, can be calculated via

$$\mathbf{D} = \sum_{i=1}^{b}\|\mathcal{A}(:,:,i)\|^2. \tag{22}$$

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Finally, $\mathbf{D}$ is normalized as follows to obtain the final anomaly detection map:

$$\bar{\mathbf{D}} = \frac{\mathbf{D} - \min(\mathbf{D})}{\max(\mathbf{D}) - \min(\mathbf{D})}. \tag{23}$$

The main steps of the proposed DLRPSAD algorithm is given in Algorithm 1, where μ_0 is the initial value of μ and $k_{\max}$ is the maximum number of iterations given.

IV. Simulation Results

This section gives several experiments on six real hyperspectral datasets to demonstrate the hyperspectral anomaly detection performance of the proposed DLRPSAD method. Some introductions about these six datasets are given below. Several classical methods are introduced as comparisons to demonstrate the effectiveness of the proposed DLRPSAD method. All the experiments are carried out on a personal computer with an Intel Core i9-12900K 3.20-GHz central processing unit and an RTX 3090Ti GPU, with 64-GB memory and 64-bit Windows 10. The model-driven methods are carried out on MATLAB 2022a, and the data-driven and proposed methods are implemented based on Pytorch 1.13.0.

A. Hyperspectral Datasets

The first dataset, beach-3, was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over a beach area of Bay Champagne. This dataset has 100 × 100 pixels, 188 effective bands, with the spectral resolution being 17.2-m. A ship in the sea consisting of 11 pixels is taken as the anomaly. This is a simple scene with a relatively simple background and a small number of small anomaly detection. The second dataset, HYDICE, was captured by the Hyperspectral Digital Imagery Collection Experiment. This dataset has 80 × 100 pixels and 175 effective bands. There are nine anomalies consisting of 21 pixels in this scene. This scene is a typical small target detection scene. The third dataset, Pavia, was captured by Reflective Optics System ImTaging Spectrometer (ROSIS-03) 466 sensor over Pavia, Italy. This dataset has 100 × 100 pixels, 102 effective bands, with the spectral resolution being 1.3-m. In this scene, the cars on the bridge with 71 pixels are considered anomalies. The fourth dataset, urban-3, was captured by AVIRIS sensors over Gainesville. This dataset has 180 × 180 pixels, 120 effective bands, and 3.5-m spectral resolution. The anomalies are composed of 52 pixels in this scene. This dataset is a typical small and dense target detection scene. The fifth dataset, airport-2, was captured by AVIRIS sensors over Los Angles. This dataset has 80 × 100 pixels, 175 effective bands, and 7.1-m spectral resolution. Two aircraft containing 87 pixels are taken as anomalies in the scene. The sixth dataset, airport-4, was captured by AVIRIS sensors over Gulfport. This dataset has 100 × 100 pixels, 191 effective bands, and 3.4-m spectral resolution. One big and two small aircraft containing 60 pixels are taken as anomalies in this scene. The pseudocolor images and their corresponding anomaly maps are shown in the first and second columns of Fig. 2.

B. Evaluation Indicators and Comparison Methods

1) In this article, we adopt five more common metrics to evaluate the hyperspectral anomaly detection performance of different methods. The first one is the anomaly detection map. Different colors in the map indicate different detection values of pixels. The second one is the receiver operating characteristics (ROC) curve. Specifically, the 2D-ROC curve plotted by the probability of detection (PD) and the probability of false alarm (PF) and the 3D-ROC curve plotted by the detection threshold τ, PD and PF, are adopted in this article. The third one is the area under the ROC curve (AUC). There are three different AUC values associated with the PD, PF, and τ, i.e., $\mathrm{AUC}_{(\mathrm{PD,PF})}$, $\mathrm{AUC}_{(\mathrm{PD},\tau)}$, and $\mathrm{AUC}_{(\mathrm{PF},\tau)}$ as described in [66]. The larger the values of $\mathrm{AUC}_{(\mathrm{PD,PF})}$ and $\mathrm{AUC}_{(\mathrm{PD},\tau)}$, the better the anomaly detection performance. While the smaller the value of $\mathrm{AUC}_{(\mathrm{PF},\tau)}$ means better background suppression performance. Combining $\mathrm{AUC}_{(\mathrm{PD,PF})}$, $\mathrm{AUC}_{(\mathrm{PD},\tau)}$, and $\mathrm{AUC}_{(\mathrm{PF},\tau)}$, the overall detection probability denoted as $\mathrm{AUC}_{(\mathrm{ODP})}$, is introduced to evaluate anomaly detection performance. The $\mathrm{AUC}_{(\mathrm{ODP})}$ is defined as

$$\mathrm{AUC}_{(\mathrm{ODP})} = \mathrm{AUC}_{(\mathrm{PD,PF})} + \mathrm{AUC}_{(\mathrm{PD},\tau)} - \mathrm{AUC}_{(\mathrm{PF},\tau)}. \tag{24}$$

Besides, the signal-to-noise probability ratio (SNPR) denoted by $\mathrm{AUC}_{(\mathrm{SNPR})}$ is also adopted to evaluate the detection performance, which is the most effective detection measure. The $\mathrm{AUC}_{(\mathrm{SNPR})}$ is defined as

$$\mathrm{AUC}_{(\mathrm{SNPR})} = \frac{\mathrm{AUC}_{(\mathrm{PD},\tau)}}{\mathrm{AUC}_{(\mathrm{PF},\tau)}}. \tag{25}$$

The fourth one is the background-anomaly separation map. The separation map reflects the statistical characteristics of abnormal and background pixels in the detection result. The last one is the running time of different methods, which is adopted to evaluate the computational complexity of the different methods.

2) Ten classical methods are introduced as a comparison to verify the effectiveness of the proposed DLRPSAD method, including two classical methods called RX in [12] and LRX in [67]; two matrix-based methods called GTVLRR in [42], and LARSR in [39]; four tensor-based methods called PTA in [47], LARTVAD in [51], PCA-TLRSR in [49], and TRDFTVAD in [52]; two AE-based methods called auto-AD in [57] and DeepLR in [62]. In constructing the background dictionary in LRASR, GTVLRR, and LARTVAD, the number of clusters is set to 15, and 20 pixels are selected as atoms in each cluster. The parameters λ and β in LRASR are chosen from the set $\{0.001, 0.01, 0.1, 0.5, 1, 2, 3\}$ and $\{0.001, 0.01, 0.05, 0.1, 0.5, 1\}$. In GTVLRR, the values of the three parameters λ, β, and γ, are chosen from the set $\{0.005, 0.01, 0.1, 0.2, 0.3, 0.5, 1, 2, 3\}$. In PTA, the value of the parameter r is selected from 0 to 20, and the value of the parameter μ is selected from the set $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$.

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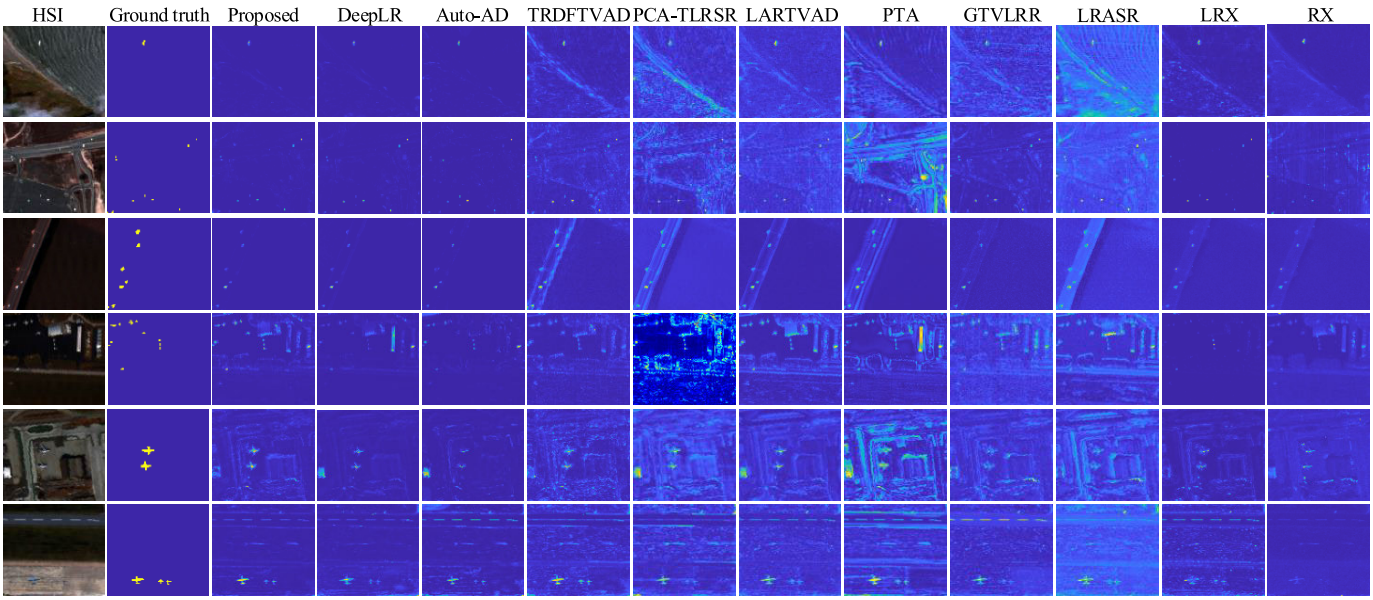


Fig. 2. Anomaly detection maps obtained by different methods on the six datasets. From top to bottom: beach-3, HYDICE, Pavia, urban-3, airport-2, and airport-4.

TABLE II
AUC VALUES OF DIFFERENT METHODS ON THE SIX DATASETS

Dataset		Proposed	DeepLR	Auto-AD	TRDFTVAD	PCA-TLRSR	LARTVAD	PTA	GTVLRR	LARSR	LRX	RX
beach-3	$AUC_{(PD,PF)}$	**0.9998**	0.9983	0.9984	0.9960	0.9749	0.9994	0.9983	0.9916	0.9374	0.9943	0.9998
	$AUC_{(PD,\tau)}$	0.2854	0.2740	0.3811	0.5257	0.5008	0.6590	0.5882	0.5349	0.5694	0.4532	0.5314
	$AUC_{(PF,\tau)}$	0.0036	0.0040	0.0035	0.0379	0.1084	0.0743	0.0670	0.0999	0.2241	0.0232	0.0259
	$AUC_{(ODP)}$	1.2817	1.2683	1.3760	1.4838	1.3674	**1.5842**	1.5195	1.4266	1.2827	1.4244	1.5053
	$AUC_{(SNPR)}$	79.9674	68.8203	**108.9026**	13.8881	4.6209	8.8750	8.7765	5.3540	2.5413	19.5493	20.4912
HYDICE	$AUC_{(PD,PF)}$	**0.9997**	0.9940	0.9854	0.9985	0.9435	0.9971	0.8713	0.9953	0.9972	0.9917	0.9857
	$AUC_{(PD,\tau)}$	0.3308	0.3446	0.2555	0.5293	0.4842	0.5316	0.4682	0.4583	0.5552	0.2612	0.2404
	$AUC_{(PF,\tau)}$	0.0051	0.0059	0.0049	0.0454	0.0866	0.0910	0.1903	0.0335	0.1076	0.0026	0.0351
	$AUC_{(ODP)}$	1.3254	1.3327	1.2359	**1.4824**	1.3411	1.4377	1.1492	1.4200	1.4448	1.2504	1.1910
	$AUC_{(SNPR)}$	65.0108	58.5166	52.0885	11.6689	5.5916	5.8393	2.4605	13.6617	5.1602	**101.9744**	6.8442
Pavia	$AUC_{(PD,PF)}$	**0.9990**	0.9912	0.9965	0.9960	0.9856	0.9970	0.9950	0.9666	0.9966	0.9942	0.9934
	$AUC_{(PD,\tau)}$	0.1221	0.1228	0.1037	0.4239	0.4111	0.3182	0.4024	0.1566	0.4442	0.1317	0.1600
	$AUC_{(PF,\tau)}$	0.0009	0.0023	0.0011	0.0317	0.0652	0.0178	0.0249	0.0224	0.0826	0.0018	0.0258
	$AUC_{(ODP)}$	1.1201	1.1116	1.0992	**1.3882**	1.3355	1.2974	1.3725	1.1007	1.3582	1.1241	1.1277
	$AUC_{(SNPR)}$	**125.2775**	52.5836	98.2756	13.3857	6.3034	17.8969	16.1415	6.9822	5.3753	72.4428	6.2103
urban-3	$AUC_{(PD,PF)}$	**0.9911**	0.9750	0.9734	0.9881	0.8968	0.9867	0.9693	0.9825	0.9617	0.9628	0.9513
	$AUC_{(PD,\tau)}$	0.3791	0.0900	0.1773	0.2227	0.3543	0.4351	0.3288	0.4338	0.3563	0.0784	0.0963
	$AUC_{(PF,\tau)}$	0.0169	0.0101	0.0064	0.0342	0.1144	0.0599	0.0604	0.1074	0.0974	0.0035	0.0351
	$AUC_{(ODP)}$	1.3533	1.0548	1.1444	1.1766	1.1367	**1.3619**	1.2376	1.3089	1.2207	1.0376	1.0125
	$AUC_{(SNPR)}$	**22.4627**	8.8720	27.7583	6.5083	3.0964	7.2642	5.4387	4.0401	3.6592	22.1880	2.7451
airport-2	$AUC_{(PD,PF)}$	**0.9818**	0.9690	0.8754	0.9759	0.9668	0.9656	0.9363	0.9361	0.8869	0.9472	0.8404
	$AUC_{(PD,\tau)}$	0.3297	0.1823	0.0959	0.2768	0.4257	0.3419	0.4732	0.2532	0.2877	0.1850	0.1841
	$AUC_{(PF,\tau)}$	0.0287	0.0136	0.0119	0.0573	0.1126	0.0702	0.1389	0.0813	0.1266	0.0243	0.0516
	$AUC_{(ODP)}$	**1.2828**	1.1377	0.9593	1.1954	1.2799	1.2373	1.2705	1.1079	1.0480	1.1079	0.9729
	$AUC_{(SNPR)}$	11.4881	**13.3766**	8.0503	4.8309	3.7809	4.8683	3.4062	3.1135	2.2727	7.6269	3.5691
airport-4	$AUC_{(PD,PF)}$	**0.9980**	0.9904	0.9785	0.9978	0.9804	0.9938	0.9922	0.9956	0.9885	0.9599	0.9526
	$AUC_{(PD,\tau)}$	0.4760	0.3495	0.3294	0.5979	0.4566	0.4906	0.6687	0.4191	0.4350	0.2347	0.0736
	$AUC_{(PF,\tau)}$	0.0120	0.0216	0.0248	0.0534	0.0747	0.0687	0.0957	0.0561	0.1017	0.0472	0.0248
	$AUC_{(ODP)}$	1.4621	1.3183	1.2831	1.5423	1.3622	1.4157	**1.5652**	1.3586	1.3218	1.1473	1.0015
	$AUC_{(SNPR)}$	**39.7321**	16.2089	13.2821	11.1949	6.1086	7.1457	6.9857	7.4685	4.2762	4.9713	2.9743

Other parameters of the PTA algorithm remain unchanged as in [50]. For the parameters λ and β in LARTVAD, the values of those two parameters are chosen from the set $\{0.01, 0.1, 1, 2, 3, 10\}$. Other parameters remain unchanged. The parameters λ and λ' in PCA-TLRSR are chosen from the set $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$, one of them is adjusted with the other being fixed. Other parameters remain unchanged. The parameters λ and β in TRDFTVAD are chosen from the set $\{0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$, the selection range of parameters r_1 and r_2 is the same as in [52], and other parameters remain unchanged. The value of the parameter σ in auto-AD is set the same as that in [57].

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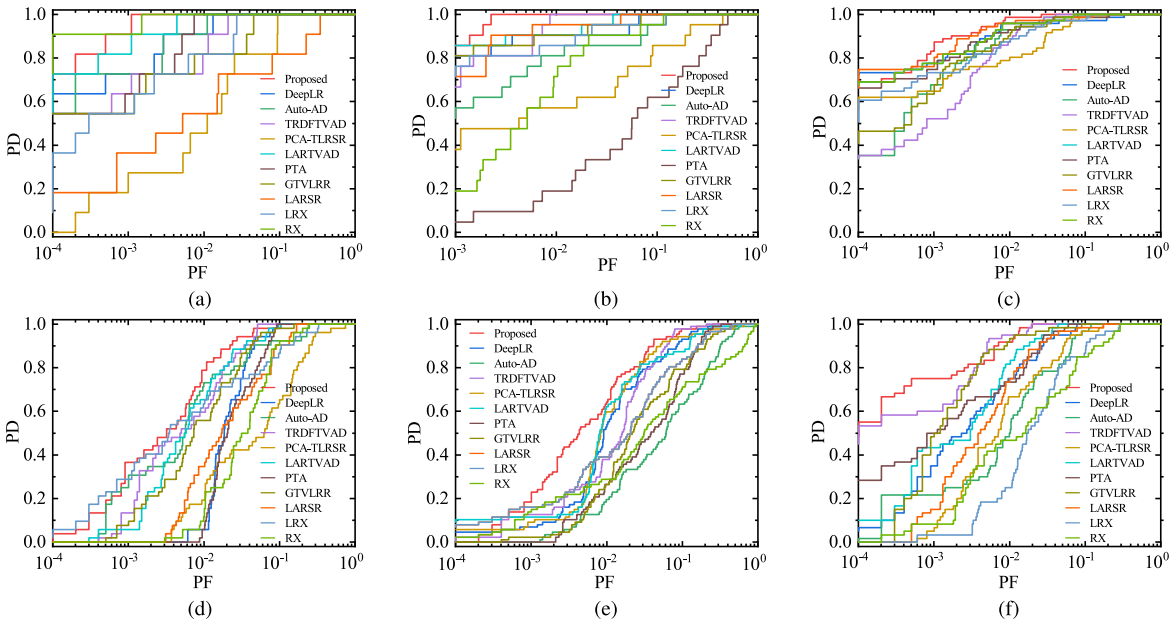


Fig. 3. Two-dimensional-ROC curve of the proposed method and the compared methods. (a) Beach-3 dataset. (b) HYDICE dataset. (c) Pavia dataset. (d) Urban-3 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

The values of all parameters in DeepLR are set the same as those in [62].

C. Experiment Results and Analysis

Fig. 2 shows the detection maps obtained by different methods on the six datasets to more intuitively evaluate the hyperspectral anomaly performance of the proposed DLRP-SAD method. For the beach-3 dataset shown in the first row in Fig. 2, the proposed method, the DeepLR and the auto-AD, seem to have the same background suppression performance, better than other comparison methods. Meanwhile, the anomaly detection performance of those three methods is also good. The discrimination of the background pixels and the anomaly pixels in detection maps obtained by the proposed method, the DeepLR and the auto-AD, are the same, better than other comparison methods. For the HYDICE dataset shown in the second row in Fig. 2, the LRX seems to have the best background suppression performance. The proposed method, the DeepLR, and the auto-AD seem to have the same background suppression performance, better than other comparison methods except for the LRX. The discrimination of the background pixels and the anomaly pixels in the detection map obtained by the LRX is better than other comparison methods. The anomaly detection performance of the proposed method seems to be the same as the DeepLR and the auto-AD. For the Pavia dataset shown in the third row in Fig. 2, the proposed method has the best background suppression performance, and the anomaly detection performance of the proposed method is also good. The discrimination of the background pixels and the anomaly pixels in the detection map obtained by the PTA is better than other comparison methods. For the urban-3 dataset shown in the fourth row in Fig. 2, the LRX has the best background suppression performance but misses many anomaly pixels. The proposed method has the best anomaly detection performance, and discrimination of the background pixels and the anomaly pixels in the detection map obtained by the LRX is better than other comparison methods. For the airport-2 dataset shown in the fifth row in Fig. 2, the auto-AD has the best background suppression performance but misses some anomaly pixels. The proposed method has the best anomaly detection performance, and the discrimination of the background pixels and the anomaly pixels in detection maps obtained by the proposed methods is better than other comparison methods. For the airport-4 dataset shown in the last row in Fig. 2, the RX has the best background suppression performance but misses many anomaly pixels. The discrimination between the background pixels and the anomaly pixels in the detection map obtained by the TRDFTVAD is better than other comparison methods. The anomaly detection and background suppression performance of the proposed method are also relatively good.

Fig. 3 gives the 2D-ROC curve obtained by different methods for the six datasets. For the HYDICE and airport-4 datasets, for almost every value of PF alarm, the PD of the proposed method is the highest among all tested detectors, as we can see from Fig. 3(b) and (f). For the Pavia, urban-3, and airport-2 datasets, the PD of the proposed method is higher than other comparison methods when PF $> 10^{-3}$, as we can see from Fig. 3(c)–(e). For the beach-3 dataset, the PD of the proposed method is higher than other comparison methods only in some intervals of PF, as we can see from Fig. 3(a). The 3D-ROC curve obtained by different methods for the six datasets are displayed in Fig. 4. Combining the information from Figs. 3 and 4, we can know that the proposed algorithm has a good anomaly detection performance on the six datasets. In addition, the values of the five types of AUC for all comparison methods are given in the Table II to quantitatively

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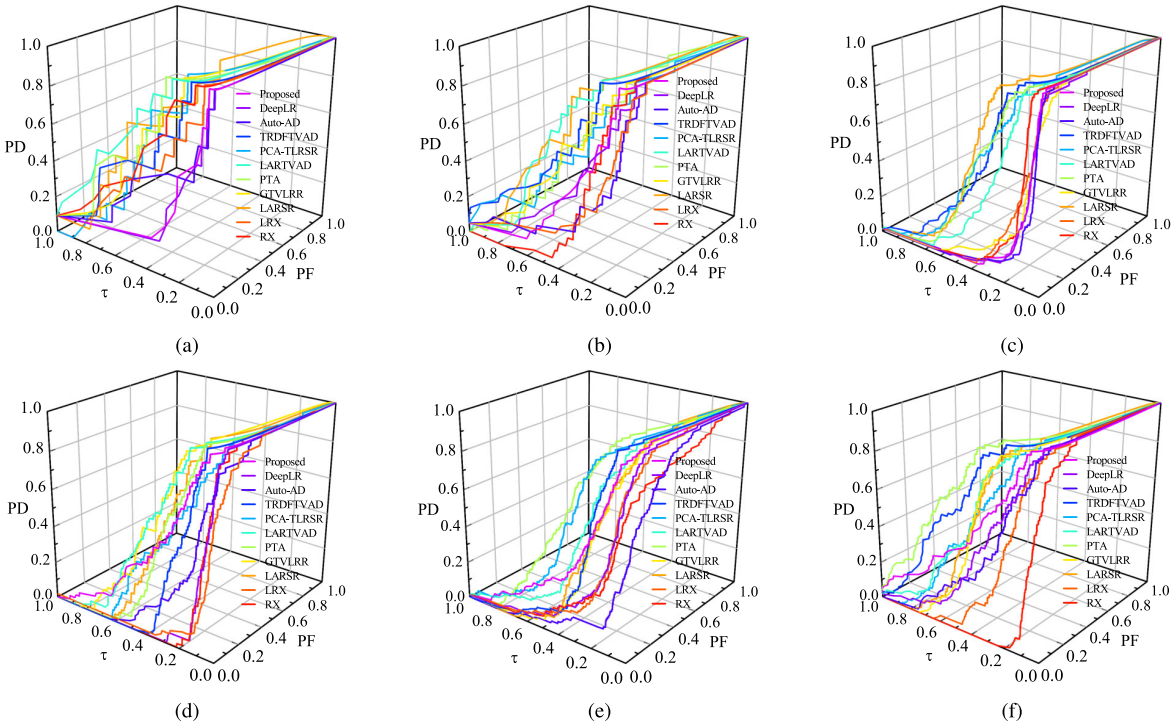


Fig. 4. Three-dimensional-ROC curve of the proposed method and the compared methods. (a) Beach-3 dataset. (b) HYDICE dataset. (c) Pavia dataset. (d) Urban-3 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

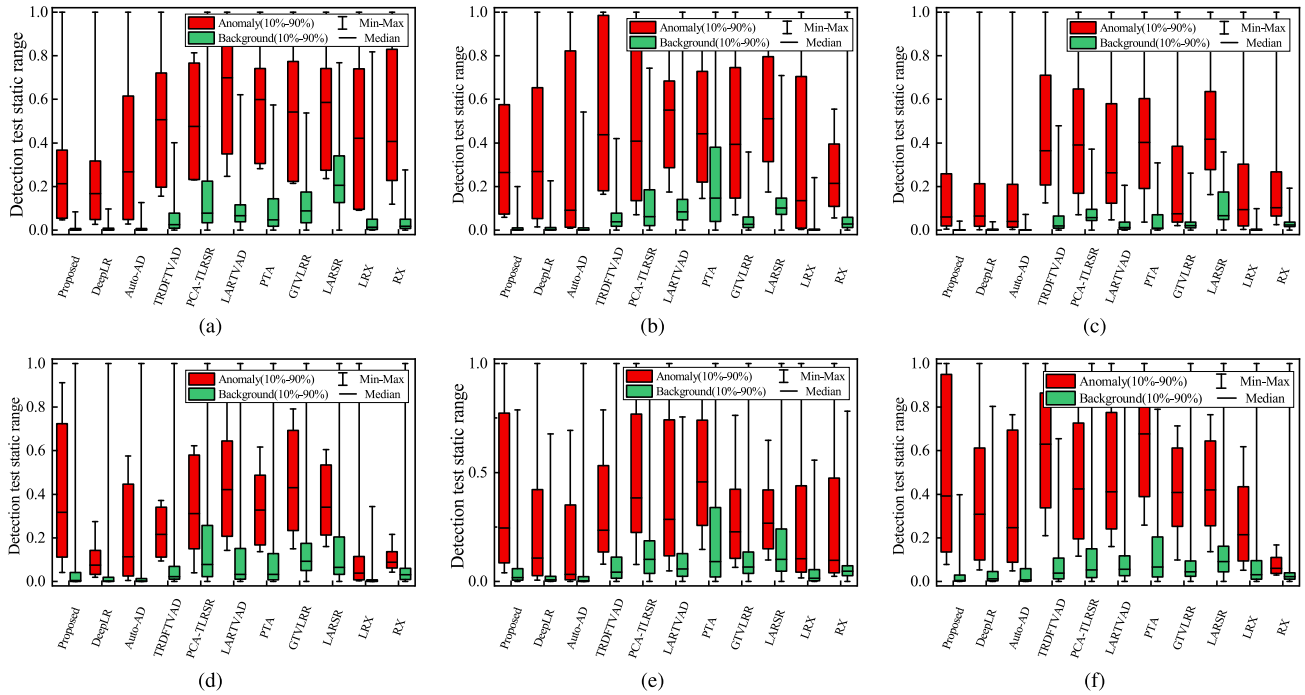


Fig. 5. Background-anomaly separation maps of the proposed method and the comparison methods. (a) Beach-3 dataset. (b) HYDICE dataset. (c) Pavia dataset. (d) Urban-3 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

compare the hyperspectral anomaly detection performance of these methods on the six datasets. The $AUC_{(PD,PF)}$, $AUC_{(ODP)}$, and $AUC_{(SNPR)}$ are selected as the final evaluation indicators among the five types of AUC. The values of $AUC_{(PD,PF)}$ of the proposed method on the six datasets are the highest among all the comparison methods. For Pavia, urban-3, and airport-4 datasets, the proposed methods has the best $AUC_{(SNPR)}$ value. For airport-2 dataset, the proposed methods has the best $AUC_{(ODP)}$ value. The best values of the three indicators are marked in black in Table II. Thus, we can know that the proposed method has an excellent anomaly detection performance.

The background-anomaly separation maps of the six datasets are displayed in Fig. 5, in which the red and green boxes represent the pixels of anomaly and background distributed in the range of 10%–90%, respectively. The horizontal line in each box denotes the median value of the pixels. The vertical bars above and below the box

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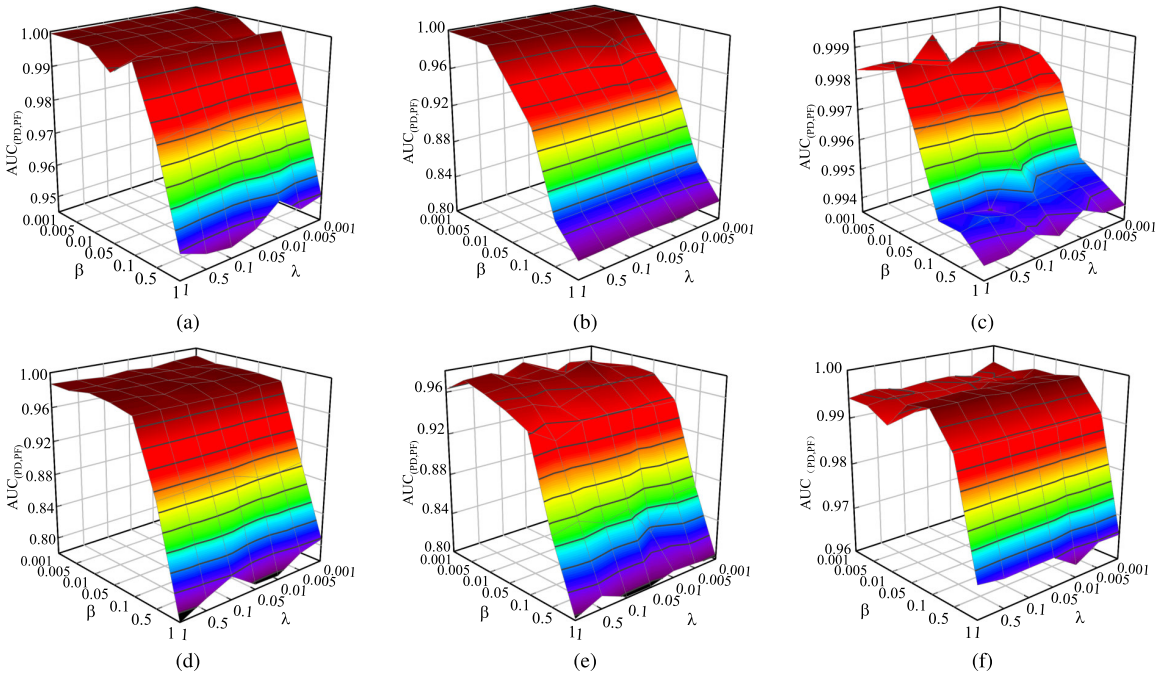


Fig. 6. $AUC_{(PD,PF)}$ values via the parameters λ and β of the proposed method in the six datasets. (a) Beach-3 dataset. (b) HYDICE dataset. (c) Pavia dataset. (d) Urban-3 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

TABLE III
RUNNING TIME OF DIFFERENT ALGORITHMS ON THE SIX DATASETS

Dataset	Proposed	DeepLR	Auto-AD	TRDFTVAD	PCA-TLRSR	LARTVAD	PTA	GTVLRR	LARSR	LRX	RX
beach-3	77.3480	4.8699	5.4930	46.1620	11.6000	19.6150	12.8240	45.2320	11.1170	6.6290	0.1830
HYDICE	50.8601	4.0624	13.0729	44.1290	8.6240	36.4340	11.0310	39.7850	10.9240	4.8280	0.1800
Pavia	42.6249	3.2906	7.1032	23.5310	4.5150	21.5450	8.5840	39.2590	11.3640	2.7200	0.1700
urban-3	104.4460	5.0042	5.8380	52.2690	5.7460	32.2910	11.6660	41.4420	11.9560	6.7270	0.1870
airport-2	80.9392	5.3720	4.9673	38.9350	10.2020	44.1006	12.7020	44.8080	13.5490	7.4950	0.1860
airport-4	155.5049	4.9018	4.0748	59.1650	10.8210	33.0010	11.6450	43.8780	13.3450	6.8700	0.1860

denote the pixels distributed in the range of 90%–100% and 0%–10%, respectively. For the beach-3 dataset shown in Fig. 5(a), the LARTVAD gets the best separation performance, the LARSR seems to have the worst separation performance. The separation performance of the proposed method is only slightly better than that of the DeepLR and auto-AD. For the HYDICE dataset displayed in Fig. 5(b), the best separation performance is won by LARSR, and the PTA gets the worst separation performance. The separation performance of the proposed method is better than that of the DeepLR and the auto-AD. For the Pavia dataset displayed in Fig. 5(c), the TRDFTVAD gets the best separation performance. The separation performance of the proposed method is close to that of the DeepLR, better than that of the auto-AD. For the urban-3 dataset shown in Fig. 5(d), the proposed method gets the best separation performance, and the PCA-TLRSR gets the worst separation performance. The TRDFTVAD and LARTVAD seem to have the same performance, and all are better than other comparison methods. For the airport-2 dataset displayed in Fig. 5(e), the proposed method has the same separation performance as the TRDFTVAD and PCA-TLRSR, better than the other comparison methods. For the airport-4 dataset displayed in Fig. 5(f), the TRDFTVAD has the best separation performance. The separation performance of the proposed method is better than the DeepLR and auto-AD. Overall, the proposed DLRPSAD model shows an excellent background-anomaly separation performance on the six datasets.

Table III lists the running time of the different algorithms, which is adapted to compare the computational complexity of different algorithms. It can be seen from Table III that the running time of the RX algorithm is the most economical, but its anomaly detection performance is not good. Compared with DeepLR, the computational time of the proposed algorithm takes longer since the proposed method needs to perform multi-times singular value decomposition operation when solving the parameter $\mathcal{X}$ in (8), which is a time-consuming process, but the performance of the proposed algorithm is better than DeepLR, which is understandable.

D. Parameters Setting

In the proposed DLRPSAD model, two parameters λ and β, need to be artificially designed to optimize the performance of the proposed method. The parameters λ and β determine the weights of the TNN and 3DTV regularization on the reconstructed background tensor $f_\theta(\mathcal{Y})$, respectively. To test the impact of different values of λ and β on the

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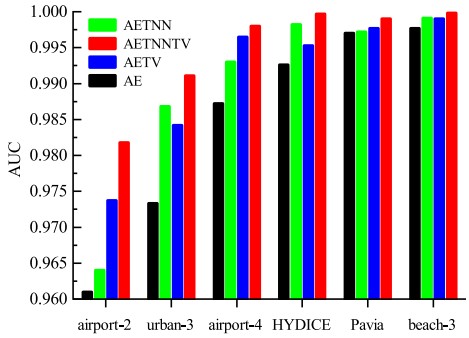


Fig. 7. $AUC_{(PD,PF)}$ values of the AE detector with different regularizations on the six datasets.

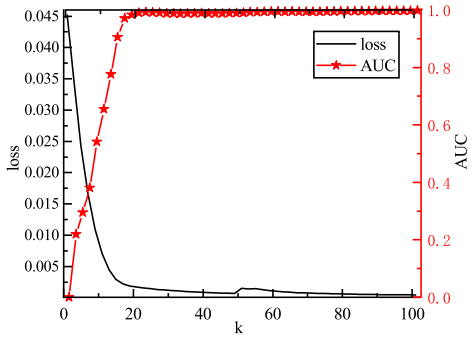


Fig. 8. Loss caves and $AUC_{(PD,PF)}$ values during the different number of iterations on the beach-3 dataset.

anomaly detection performance and to simplify the experimental analysis at the same time, the values of λ and β are all selected from $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1\}$ in this article. The $AUC_{(PD,PF)}$ value is adopted to evaluate the detection performance under different λ and β values. The values of $AUC_{(PD,PF)}$ on the six datasets under the various values of (λ, β) are shown in Fig. 6. For the beach-3 dataset shown in Fig. 6(a), the values of $AUC_{(PD,PF)}$ change little when $\beta < 0.5$, the value of $AUC_{(PD,PF)}$ hardly changes with the change of the value of λ. We get the best $AUC_{(PD,PF)}$ value when $\lambda = 0.001$ and $\beta = 0.001$. For the HYDICE dataset shown in Fig. 6(b), the values of $AUC_{(PD,PF)}$ change little when $\beta < 0.05$, the value of $AUC_{(PD,PF)}$ hardly changes with the change of the value of λ. We get the best $AUC_{(PD,PF)}$ value when $\lambda = 0.001$ and $\beta = 0.01$. For the Pavia dataset shown in Fig. 6(c), generally speaking, the value of $AUC_{(PD,PF)}$ varies little with the change of the values of λ and β, the value of $AUC_{(PD,PF)}$ decreases slightly when the value of β is greater than 0.01. We can get the best $AUC_{(PD,PF)}$ value when $\lambda = 0.05$ and $\beta = 0.001$. For the urban-3 dataset shown in Fig. 6(d), the value of $AUC_{(PD,PF)}$ varies little when $\beta < 0.5$, and hardly changes with the change of the value of λ. We get the best $AUC_{(PD,PF)}$ value when $\lambda = 0.005$ and $\beta = 0.01$. For the airport-2 dataset shown in Fig. 6(e), the value of $AUC_{(PD,PF)}$ change a little when $\beta < 0.05$, and hardly changed with the change of the value of β. We get the best $AUC_{(PD,PF)}$ value when $\lambda = 0.01$ and $\beta = 0.01$. For the airport-4 dataset shown in Fig. 6(f), we can get a good $AUC_{(PD,PF)}$ value when $\beta < 1$, and we get the best $AUC_{(PD,PF)}$ value when $\lambda = 0.05$ and $\beta = 0.1$. The above simulation results show that when β is small, the proposed model is more robust to parameters λ and β for most datasets. Of course, for different datasets, it is still necessary to manually set the parameters to achieve the optimal performance of the proposed method.

E. Ablation Study

To better demonstrate the effectiveness of the proposed algorithm, this section conducted the following two ablation experiments. One is to verify the effectiveness of designing the TNN and 3DTV regularizations on the output of the AE network, and the other is to demonstrate the convergence of the proposed algorithm.

First, we conducted comparative experiments on the six datasets to compare the anomaly detection results obtained by the following detectors: the detector of the pure AE network (marked as AE), the detector of the AE network with TNN/3DTV regularization (marked as AETNN/AETV), and the detector of the AE network with both TNN and 3DTV regularizations (marked as AETNNTV). To simplify the experiment, the values of the TNN regularization and 3DTV regularization weight coefficients λ and β are the same as the values when the AUC achieves the optimal value in the above simulation. The anomaly detection results are displayed in Fig. 7. It can be seen from Fig. 7 that designing TNN or 3DTV regularization on the output of the AE network improves the anomaly detection performance to a certain extent compared with the AE without those regularizations. Both the TNN and 3DTV regularization positively impact the detection results. Meanwhile, the anomaly detection performance of the AE network with both TNN and 3DTV regularizations is better than that of the AE network with only TNN or TV regularization since this detector simultaneously exploits the low-rank and piecewise-smooth properties of the background in three dimensions, thereby achieving better anomaly detection performance. From Fig. 7, we can also see that the reconstructed background does have low-rank and piecewise-smooth properties in the spatial and spectral dimensions, so it is effective to design tensor low-rank and 3DTV regularizations on the output of the AE network.

Second, we tested the convergence of the proposed algorithm on the beach-3 dataset using the loss curve and the values of $AUC_{(PD,PF)}$, the results are displayed in Fig. 8. It can be seen from Fig. 8 that the loss of the algorithm generally decreases with the increase of the number of iterations and finally tends to converge. It is noticed that the loss increases slightly when the number of iterations is 50 because the weight $\mathcal{M}$ in (16) is updated at this time, and the loss is calculated with this new weight. However, the loss trend decreases with the same weight $\mathcal{M}$. In addition, it can also be seen from the Fig. 8 that the value of $AUC_{(PD,PF)}$ gradually increases as the number of iterations increases and eventually tends to a stable value. Thus, it can be seen that the proposed algorithm is convergent from the above analysis.

V. Conclusion

In this article, we proposed a novel deep low-rank and piecewise-smooth constraint tensor model for hyperspectral anomaly detection. The proposed method reconstructs the background tensor through an AE network with adaptive

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weights. The input and output of the network remain in the 3-D cube shape of the raw hyperspectral, thus protecting the multidimensional structure of the data. The TNN and 3DTV regularization are designed on the output of the network to exploit the low-rank and piecewise-smooth properties of the background in the 3-D. An efficient solution scheme based on the ADMMs is proposed to divide the network train and the low-rank and piecewise-smooth background tensor reconstruct into several independent sub-problems, making the solving process more accessible. The proposed method has excellent hyperspectral anomaly detection performance since it combines the advantages of the data-driven and model-driven methods. Experiments on six real datasets demonstrate the effectiveness of the proposed method.

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